

Towards Abstract Meaning Representation for Konkani

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Abstract

Abstract Meaning Representation (AMR) provides a language-agnostic, graph-based encoding of sentence meaning that enables high-level semantic processing for downstream NLP tasks. Despite growing multilingual AMR research, most existing datasets and parsers target high-resource languages. We introduce the first AMR dataset for Konkani language. The dataset contains 1,100 sentence-graph pairs. We benchmark three parsing strategies: zero-shot transfer from a multilingual AMR model, supervised fine-tuning and language-adaptive pretraining on Konkani language data. We find that finetuning increases the Smatch F1 (0.206 $\rightarrow$ 0.331), and language-adaptive pretraining improves it further to 0.437, demonstrating that even modest annotation and domain-specific pretraining yield meaningful gains for low-resource semantic parsing. Dataset and code are available at <https://github.com/aatmanvaidya/konkani-amr>.

1 Introduction

Abstract Meaning Representation (AMR) (Banarescu et al., 2013) is a graph-based semantic formalism that encodes the meaning of a sentence as a rooted, directed, acyclic graph. Nodes represent concepts, while edges encode semantic relations such as predicate-argument structure, modifiers, and named entities. AMR has become an important resource for downstream tasks including question answering, machine translation, and information extraction, because it captures *who did what to whom* in a language-agnostic way that abstracts over surface syntactic variation (Cai and Lam, 2020; Zhang et al., 2019). Despite sustained progress in AMR research, virtually all existing datasets and parsers target a small set of high-resource languages primarily English, with recent extensions to Chinese, Spanish, German, and a handful of other well-resourced languages (Vanroy and Van de Cruys, 2024; Cai et al., 2021).

Extending AMR parsing to low-resource languages remains a major challenge in NLP. Most advances in semantic representation learning rely on the availability of large annotated corpora, parallel data, and linguistic resources that are unavailable for low-resource languages. As a result, low-resource languages are often excluded from the development of semantic parsing systems, limiting the applicability of meaning-based NLP technologies across diverse linguistic communities (Oral et al., 2024; Mansouri, 2025; Barta et al., 2025; Martinez et al., 2024). This disparity is particularly evident in the Indian linguistic landscape. NLP research has primarily focused on a small number of high-resource languages such as Hindi, Bengali, and Tamil (Gala et al., 2023). Many regional languages continue to lack annotated datasets, semantic resources, and pretrained language technologies. The scarcity of such resources makes it difficult to develop robust semantic parsers and to evaluate cross-lingual transfer methods in truly low-resource settings.

Konkani presents a compelling and underexplored case. It is the official language of the Indian state of Goa, despite its official constitutional status, Konkani is severely under-resourced in NLP; it lacks large annotated corpora, robust parsers, and existing models for neighbouring languages such as Hindi or Marathi do not transfer well because of Konkani’s distinct morphology and lexicon (see Section 2). This paper presents the first AMR dataset for Konkani, comprising 1,100 sentences drawn from a parallel translation corpus and a local Konkani newspaper. We also do a systematic comparison of three parsing pipelines, zero-shot multilingual AMR transfer, supervised fine-tuning, and language-adaptive pretraining followed by AMR fine-tuning. Fine-tuning on our dataset raises the Smatch F1 from 0.206 to 0.331. Language-adaptive pretraining on 189,332 unlabelled Konkani sentences further improves the

score to 0.437, demonstrating a clear and consistent benefit from each additional adaptation step. Together, these results show that modest annotation investment combined with in-domain pre-training can produce increasingly capable low-resource AMR parsers.

The remainder of the paper is organised as follows. Section 2 situates Konkani as a low-resource language and surveys existing linguistic resources. Section 3 reviews prior work on AMR parsing, multilingual extension, and language-adaptive pre-training. Section 4 describes the collection and annotation of the dataset. Section 5 presents our three experimental systems. Section 6 reports quantitative and qualitative results. Section 7 discusses findings, limitations, and future directions.

2 Background

Socio-cultural and Linguistic Profile. Konkani is an Indo-Aryan language spoken by the Konkani people, primarily in the Konkani region, along the western coast of India.¹ It is one of the 22 scheduled languages mentioned in the Indian Constitution. It is the sole official language of the state of Goa and is also spoken in the coastal districts of Karnataka, Kerala, and Maharashtra. The 2011 Census of India recorded approximately 5.9 million speakers of the language. Konkani has historically been overshadowed by Marathi (the dominant language of the surrounding region) and Portuguese (the colonial language of Goa for over 450 years). Konkani is characterised by different dialectal variations. Major dialects include Antruzi, Bardezi, Saxtti, and Malvani, among others. Written Konkani appears in multiple scripts: Devanagari (the official and most widely used script in Goa), Kannada script (in Karnataka), Malayalam script (in Kerala), and the Roman script (historically used by the Goan Catholic community)². Linguistically, Konkani is a morphologically rich language with a relatively free word order (predominantly SOV) (Desai et al., 2014; Martínez de Antoñana, 2018). Its nominal morphology encodes gender (masculine, feminine, neuter), number, and case. Verbal morphology is complex, encoding tense, aspect, mood, and agreement with the subject and, in some constructions, the object. These properties

pose significant challenges for tokenisation and for learning syntactic and semantic structure from limited data (Rajan et al., 2020).

Low-Resource Nature and NLP Challenges. Despite its constitutional recognition, Konkani is severely under-resourced by every standard metric used in the NLP community:

Corpora. Large-scale monolingual corpora do not exist for Konkani. Unlike Hindi or Bengali, which are represented in Common Crawl, Wikipedia, and numerous curated corpora (Kakwani et al., 2020; Ramesh et al., 2022). Konkani’s web footprint is also limited, Wikipedia in Konkani (gom.wikipedia.org) contains roughly 3,800 articles, much smaller than high-resource Indic languages.

Annotated data. There are no publicly available treebanks, named entity corpora, or semantic banks for Konkani. Universal Dependencies does not include a Konkani treebank. No AMR dataset exists prior to this work.

Pre-trained models. Konkani is absent from most large pre-trained multilingual models. The mBART tokenizer, which covers 50 languages, does not include Konkani; the closest available language code is Hindi (hi_IN) (Tang et al., 2020). IndicBERT (Kakwani et al., 2020), IndicBART (Dabre et al., 2022) and IndicTrans2 (Gala et al., 2023) include Konkani among the 22 scheduled languages, making them among the few exceptions.

Translation resources. The AI4Bharat BPCC dataset (Gala et al., 2023) provides parallel Konkani–English sentence pairs collected from government documents, news, and other public sources. This is the primary resource available for Konkani and is used in this work for both annotation of AMR and pretraining the model.

Hence, the landscape of Konkani NLP resources can be summarised as follows, Machine translation quality has improved with the release of IndicTrans2, which supports Konkani as a target language. Basic tasks such as language identification and script normalisation are partially addressed. However, higher-level semantic processing like dependency parsing, semantic role labelling, POS tagging, named entity recognition, relatively understudied for Konkani. Abstract Meaning Representation is particularly valuable in the low-resource setting because it provides a language-independent semantic understanding. An AMR graph for a

¹https://en.wikipedia.org/wiki/Konkani_language

²https://en.wikipedia.org/wiki/Konkani_in_the_Roman_script

Konkani sentence refers to the same PropBank-style predicate senses and semantic relations as English AMR, making cross-lingual transfer and evaluation possible.

3 Related Work

AMR parsing is the task of mapping a natural language sentence to its AMR graph, and has evolved through several paradigms since the formalism was introduced by [Banarescu et al. \(2013\)](#).

Transition and graph-based parsers. Early systems framed AMR parsing as a structured prediction problem. [Flanigan et al. \(2014\)](#) introduced the first AMR parser using a two-stage pipeline: concept identification followed by relation classification. Subsequent work proposed graph-to-sequence and graph-transduction approaches ([Lyu and Titov, 2018](#); [Zhang et al., 2019](#)), leveraging dependency trees and co-reference as auxiliary signals ([Konstas et al., 2017](#)). Smatch ([Cai and Knight, 2013](#)) is the standard metric for AMR evaluation. It computes precision, recall, and F1 over the set of AMR triples (variable, relation, value) by finding the maximum overlap between two graphs via a hill-climbing search.

Sequence-to-sequence parsers. The dominant methods today treat AMR parsing as a sequence-to-sequence task, in which the input is a token sequence and the output is a linearised AMR graph ([Binghao et al., 2024](#)). [Xu et al. \(2020\)](#) and [Bevilacqua et al. \(2021\)](#) demonstrated that pre-trained language models such as BART ([Lewis et al., 2020](#)) can learn this linearised representation effectively. These methods achieve state-of-the-art English parsing performance by training BART with a symmetrical objective covering both parsing and generation.

Multilingual and Cross-Lingual AMR Parsing.

Similar efforts have also been put to extended AMR parsing for multilingual languages ([Cai et al., 2021](#); [Vanroy and Van de Cruys, 2024](#)). [Damonte and Cohen \(2018\)](#) proposed cross-lingual AMR parsing by translating input sentences to English before parsing, establishing a straight-forward pipeline baseline. [Biloshmi et al. \(2020\)](#) (XL-AMR) developed a multilingual AMR dataset covering Italian, Spanish, German, and Chinese, and trained cross-lingual parsers that operate directly on the source language. [Sheng et al.](#)

(2021) proposed a unified multilingual AMR framework using mBART as the backbone. [Vanroy and Van de Cruys \(2024\)](#) extended a similar approach for Dutch and Spanish. Critically for our work, all existing multilingual AMR datasets target European or East Asian languages with large linguistic resources. To the best of our knowledge, no prior work addresses AMR parsing for any Indic language, let alone a low-resource Indic language such as Konkani. There have been efforts to do dependency parsing for Indian languages ([Bhat et al., 2018](#); [Bharati and Sangal, 1990](#); [Bharati et al., 2009](#)).

Language-Adaptive Pre-training. Pre-training large language models on domain or language-specific data before task-specific fine-tuning is a well-established technique for improving performance on low-resource tasks ([Gururangan et al., 2020](#)). For cross-lingual settings, continued pre-training on target-language text has been shown to improve both understanding and generation ([Chau et al., 2020](#); [Pfeiffer et al., 2021](#)). For seq2seq models in particular, [Liu et al. \(2020\)](#) (mBART) demonstrated that denoising autoencoding pre-training on 25 languages simultaneously yields strong multilingual translation models. The denoising objective where input text is corrupted by token masking, deletion, and sentence permutation teaches the model to reconstruct the original sequence, thereby acquiring syntactic and semantic knowledge from raw text. In our work, we continue pre-training mBART on Konkani text from BPCC dataset using a reconstruction objective ([Gala et al., 2023](#)). This is motivated by the observation that mBART’s original vocabulary and pre-training data do not include Konkani, and that exposure to in-domain Konkani text may improve the model’s encoding of Konkani morphology before AMR finetuning.

LLMs for AMR Parsing. Recent work has explored the use of LLMs for AMR parsing ([Ho, 2025](#)). Fine-tuning decoder-only LLMs has shown comparable performance as seq-to-seq models ([Ji and Wein, 2025](#); [Kamel et al., 2025](#); [Shen et al., 2024](#)). Studies have also investigated LLMs for annotating text into AMR Penman notation. While LLMs can reliably produce the basic AMR format and capture core predicate-argument structures, they often struggle with linguistic precision, frequently making errors in reentrancies and overall

semantic consistency (Raut et al., 2025; Ettinger et al., 2023). Nevertheless, they show promise for semi-automated annotation workflows and bootstrapping labeled data for low-resource languages through few-shot prompting. We adopt this method by using Gemini 2.5 pro to generate Penman-notation AMR graphs, followed by human verification.

4 Dataset

4.1 Data Collection

Our primary sentence source is the AI4Bharat BPCC (Bharat Parallel Corpus Collection) (Gala et al., 2023), a large-scale parallel corpus of 22 Scheduled Indian languages drawn from government documents, news, and web sources. We randomly sampled 1,000 Konkani sentences, filtering out those shorter than four words or longer than 50 words to avoid semantically trivial inputs. We supplemented the BPCC sentences with 100 sentences drawn from *Bhaangarbhuin*, a Konkani-language newspaper³. Newspaper text introduces colloquial registers helping the dataset better reflect the range of everyday Konkani usage. In total we had a final corpus of 1,100 Konkani sentences.

4.2 Silver Annotation using Gemini 2.5 Pro

We initially annotated a small set of sentences by following the English Propbank. Early in the process, we realised that manual AMR annotation is labour-intensive (Cai et al., 2023). And annotating 1100 sentences would take a very long time. We therefore follow recent work on silver AMR annotation (Ettinger et al., 2023; Li and Fowlie, 2025; Hite et al., 2026) and use Gemini 2.5 Pro⁴ as an automatic annotator, with subsequent human verification.

Each sentence was annotated via a structured prompt comprising: (1) a system instruction covering AMR fundamentals, Penman notation, and the core relation inventory (:ARG0–:ARG5, modifiers, named entities, date entities); (2) the Konkani source sentence; and (3) a request for the AMR graph only. Concept nodes are in English (e.g., approve-01, increase-01), consistent with PropBank conventions. The full prompt is listed in Appendix A.1.

Human Verification All 1,100 annotations were reviewed by the authors who have knowledge of both Konkani and the AMR PropBank rules and conventions. Each annotation was evaluated along three things: (a) does the graph correctly capture the meaning of the Konkani sentence?, (b) structural validity (is the Penman notation well-formed?), and (c) PropBank alignment (are predicate senses correctly chosen?). Upon review, 128 annotations (11.6%) were found to be incorrectly annotated and were fixed manually, yielding a gold Konkani AMR dataset of 1,100 sentence–graph pairs. The most common annotation errors fell into two categories: incorrect predicate frame selection (e.g., choosing a sense number that mismatches the argument structure implied by the Konkani clause) and incorrect argument role assignment, particularly confusing :ARG0 (agent) with :ARG1 (patient) in passive or nominative constructions.

Here are a few examples from of annotated sentences.⁵

Example 1

English Translation: The price of diesel has increased by 80 paise per liter.

```
(i / increase-01
 :ARG1 (p / price
        :poss (d / diesel))
 :ARG2 (m / monetary-quantity
        :quant 80
        :unit (p2 / paise)
        :per (l / liter)))
```

Example 2

English Translation: Congress has become very aggressive over the issue of rising inflation.

```
(b / become-01
 :ARG1 (p / political-party
        :name (n / name :op1 "Congress"))
 :ARG2 (a / aggressive
        :degree (v / very))
 :cause (i / issue-01
        :topic (i2 / inflate-01)))
```

³<https://epaper.thegoan.net/edition/bhaangarbhuin/23246>

⁴<https://ai.google.dev/gemini-api/docs/models/gemini-2.5-pro>

⁵The full dataset can be found here - https://github.com/aatmanvaidya/konkani-amr/blob/main/data/konkani_amr.csv

Example 3

English Translation: As per the central government’s order, traffic rules will have to be implemented in Goa.

```
(o / obligate-01
  :ARG1 (i / implement-01
    :ARG1 (r / rule
      :mod (t / traffic))
    :location (s / state
      :name (n / name :op1 "Goa")))
  :cause (o2 / order-01
    :ARG0 (g / government
      :mod (c / center))))
```

4.3 Pretraining Corpus

Alongside the annotated dataset, we compiled a larger unlabelled corpus of Konkani sentences for language pretraining. From the same AI4Bharat BPCC data, we extracted all available Konkani sentences and deduplicated them, yielding 189,332 unique sentences. This corpus is used solely for continued language model pretraining (Section 5) and plays no role in the annotation pipeline.

5 Methodology

We design three experimental systems to evaluate different strategies for building a Konkani AMR parser.

5.1 Base Model

All three systems share the same backbone: mbart-large-cc25-ft-amr30-en, a multilingual seq2seq model based on mBART (Liu et al., 2020) that has been fine-tuned on AMR 3.0 for English AMR parsing⁶. The model produces AMR graphs in a SPRING-style linearisation (Bevilacqua et al., 2021) that uses pointer tokens (<pointer:N>) to encode reentrancies and relational brackets (<rel> / </rel>) to delimit argument spans. We decode these into standard Penman notation via a deterministic post-processing step.

We select this model because it is to our knowledge one of the few publicly available multilingual AMR parser, trained to parse semantics from a non-English source language using mBART’s multilingual representations. We use the Hindi language code (hi_IN) as a proxy for Konkani in all experiments, as it is the closest Indic language code supported by the mBART tokeniser. We split the 1,100-sentence dataset into training, validation, and

test sets in an 80:5:15 ratio. All systems are evaluated using Smatch (Cai and Knight, 2013), which computes precision (P), recall (R), and F1 over the set of AMR triples. We report the average Smatch F1 across the 165-sentence test set.

5.2 System 1: Zero-Shot Baseline

The first system applies the pre-trained model directly to Konkani sentences without any task-specific adaptation. This setting serves as a baseline to evaluate how effectively a model trained on AMR 3.0 English data can generalize to a related but previously unseen language. No parameters are updated; we run inference on the sentences in the test set ($n = 165$).

5.3 System 2: Supervised Fine-Tuning

The second system finetunes the base model directly on our Konkani AMR dataset. The target output for each example is the gold Penman string (post-processed from the SPRING linearisation). Fine-tuning is performed with the Hugging Face Seq2SeqTrainer. The best checkpoint is selected by minimum validation loss on the validation set. The evaluation uses the 165-sentence test split. Final model training hyperparameters are listed in Appendix A.2.1

5.4 System 3: Language-Adaptive Pre-training

The third system adds a Konkani language-adaptive pre-training stage before AMR fine-tuning. We pretrain the model on Konkani language data to improve its language understanding. This two-step pipeline is motivated by the hypothesis that exposing the model to a large volume of Konkani text before task-specific training will improve its internal representations of Konkani morphology and vocabulary, leading to better AMR generation.

Step 1 Continued pre-training on Konkani. We further adapt mBART to Konkani through continued pre-training on a monolingual corpus consisting of 189,332 sentences. Since mBART is originally trained using a denoising sequence-to-sequence objective, we continue training the model with the same objective on Konkani text to improve its representation of the language. Specifically, portions of the input sequence are corrupted through span masking, and the model is trained to reconstruct the original sentence from the corrupted input. This objective enables the model to learn syn-

⁶<https://huggingface.co/BramVanroy/mbart-large-cc25-ft-amr30-en>

System	P	R	F1
Zero-shot baseline	0.174	0.282	0.206
Fine-tuned (Sys. 2)	0.329	0.351	0.331
Pretrain + FT (Sys. 3)	0.500	0.426	0.437

Table 1: Smatch scores on the 165-sentence test set (80/5/15 train/val/test split of 1,100 examples). P and R denote average Smatch precision and recall; F1 is the average F1 score.

tactic and semantic regularities of Konkani while retaining the multilingual knowledge acquired during the original pre-training phase.

During training, each sentence is tokenized using the mBART tokenizer and processed using DataCollatorForSeq2Seq from HuggingFace. The final model hyperparameters used for pretraining are listed in Appendix A.2.2. The resulting Konkani-adapted model is saved and used as the initialisation for Step 2.

Step 2 AMR fine-tuning. The Konkani-adapted mBART model is then fine-tuned on our Konkani AMR dataset using the identical hyperparameters and data split as System 2, enabling a direct comparison of the effect of language-adaptive pre-training.

6 Results

6.1 Main Results

Table 1 reports Smatch scores for all three systems on the 165-sentence test set.

Zero-shot baseline (System 1). Applying the pre-trained multilingual AMR model directly to Konkani without any adaptation achieves a Smatch F1 of 0.206. This baseline score demonstrates that the multilingual representations of a model trained on English AMR 3.0 does some cross-lingual transfer to an unseen Indic language. Among the best zero-shot outputs, the model scores as high as F1 = 0.62 on shorter, syntactically simple sentences with date entities and named-entity-heavy reorganisation events, suggesting that structured concept types (dates, organisations, persons) are particularly transferable across languages.

Supervised fine-tuning (System 2). Fine-tuning the model on our Konkani AMR dataset produces a decent improvement. The Smatch F1 rises to 0.331 indicating that fine-tuning substantially improves output structural validity. The improvement confirms that even a small silver-annotated

dataset of approximately 900 sentence-graph pairs provides a strong training signal for AMR parsing in a low-resource language. The top-scoring predictions reach individual F1 scores of 0.78, 0.68, and 0.67, indicating that the fine-tuned model can produce high-quality graphs for sentences with clear predicate-argument structure.

Language-adaptive pre-training + fine-tuning (System 3).

Continuing pre-training on the 189,332-sentence unlabeled Konkani corpus prior to AMR fine-tuning further improves performance, increasing the Smatch F1 score to 0.437. This represents a substantial improvement over both the direct zero-shot transfer baseline and the model fine-tuned without language-adaptive pre-training. The result suggests that exposing mBART to large amounts of monolingual Konkani text enables the model to better capture the lexical, syntactic, and semantic characteristics of the language before learning the downstream AMR parsing objective. More specifically, continued pre-training helps align the models multilingual representations with Konkani linguistic patterns, thereby reducing the language mismatch between the original English-centric AMR training distribution and the target language. Since the AMR fine-tuning dataset is relatively small, this intermediate adaptation stage provides the model with richer contextual knowledge of Konkani morphology, word usage, and sentence structure, which subsequently improves semantic parsing performance.

The gains are reflected in both precision and recall. Precision increases from 0.329 to 0.500, indicating that the generated AMR graphs contain more semantically accurate relations and concepts with fewer incorrect predictions. Recall improves from 0.351 to 0.426, suggesting that the model is also able to recover a larger portion of the gold semantic structure. Together, these improvements demonstrate that language-adaptive pre-training contributes not only to graph correctness but also to graph completeness. Overall, these findings highlight the importance of monolingual language adaptation for low-resource semantic parsing. Even without additional annotated AMR data, continued pre-training on unlabeled Konkani text yields measurable improvements

6.2 Qualitative Analysis

To examine what the score differences correspond to in the actual model outputs, we manually analysed the test set predication from the model. The analysis focuses on things like, are there any hallucinations in the model outputs? Is there any improved recovery of dates and named entities after fine-tuning? and failures on longer sentences with embedded events.

Table 2 shows an AMR-level comparison for one representative sentence. We include only the parts of the graph relevant to the analysis, rather than the full Penman output, so that the comparison remains readable.

The main difference between System 1 and the adapted systems is semantic grounding. The zero-shot baseline often produces outputs that resemble AMR graphs structurally but are not meaningfully connected to the Konkani sentence. This is clearest in Table 2. The gold graph contains two related events and three important named entities: Alauddin, Devagiri, and Jalaluddin. System 1 instead introduces entities from outside the sentence and does not recover the attack, plunder, or revolt structure. The output is therefore not simply incomplete; it represents a different semantic scene.

Supervised fine-tuning reduces this failure mode. System 2 recovers the attack event and identifies Alauddin as the attacker. This is a clear improvement over System 1, where the central event structure is absent. However, the prediction remains only partially correct. Devagiri is replaced with the incorrect name “Dahadur”, and the second event is analysed as defeat-01 rather than seizure of plunder for a revolt. The model therefore captures part of the predicate-argument structure but fails to preserve the complete event configuration.

System 3 also recovers the attack event and Alauddin, indicating that language-adaptive pre-training retains the main benefit of supervised fine-tuning. However, the output still contains important entity and relation errors. Devagiri is mapped to “Temple”, and the plunder event is connected to an unrelated destination, “Dharashtra”. This suggests that continued pre-training improves the model’s exposure to Konkani text, but does not fully resolve entity normalisation or the composition of multiple event relations.

A similar pattern appears in examples involving dates and quantities. In the sentence about reorganisation taking effect from 31 October 2019, Sys-

tem 1 recognises that a date is present but assigns the wrong month. System 2 and System 3 recover the day, month, and year correctly, although their main predicate does not exactly match the gold graph. This supports the observation in Section 6.1 that shorter sentences containing explicit dates and named entities are easier for the adapted systems.

The adapted systems still make substantial errors in predicate-frame selection. In the sentence about Chandigarh dropping to 66th place among clean cities in India in 2021, both System 2 and System 3 recover Chandigarh, India, the rank 66, and the year 2021. However, neither system represents the central event as a drop in ranking. System 2 predicts a receiving event, while System 3 reverses the relation between India and Chandigarh. This shows that lexical recovery alone is insufficient: a prediction may contain the correct entities and quantities while still assigning the wrong event semantics.

Language-adaptive pre-training gives clearer improvements in some temporal examples. In the sentence about various programmes happening over three days, System 1 hallucinates unrelated person and event content, while System 2 recognises the three-day duration but attaches it to an incorrect event. System 3 preserves the temporal quantity more clearly and links it to a more plausible event structure. The sentence-level Smatch score also reflects this improvement, increasing from 0.07 for System 1 and 0.17 for System 2 to 0.37 for System 3. This supports the quantitative improvement reported in Table 1.

The remaining failures are most visible in longer sentences with embedded semantic relations. In the driving-simulator example, the gold graph requires several connected meanings: evidence was found, simulator training took place, performance improved, and the improvement concerned road-sign recognition. None of the systems recover this full structure. The adapted systems capture isolated fragments, especially around training, but they do not compose these fragments into the correct nested AMR graph. This suggests that the current dataset is sufficient for learning simpler event structures, dates, quantities, and some named entities, but not yet enough for robust parsing of complex multi-event sentences.

Overall, the qualitative analysis supports the quantitative results in Table 1. Fine-tuning improves graph relevance and reduces the severe hallucination observed in the zero-shot baseline.

Output	Relevant AMR fragment
Gold	attack-01(ARG0: Alauddin, ARG1: Devagiri, time: 1296); seize-01(ARG0: Alauddin, ARG1: plunder, purpose: revolt against Jalaluddin)
System 1	No corresponding attack or revolt structure is recovered; the output introduces unrelated entities including <i>Benito Mussolini</i> and <i>Al-Qaeda</i> .
System 2	attack-01(ARG0: Alauddin, ARG1: Dahadur, time: 1296); the second event is represented as defeat-01 rather than seizure of plunder for revolt.
System 3	attack-01(ARG0: Alauddin, ARG1: Temple, time: 1296); the output includes plunder-01, but links it to an unrelated destination, “Dharashtra”.

Table 2: AMR-level comparison for a sentence with multiple named entities.

Language-adaptive pre-training provides further gains in some cases, especially for temporal and lexical information. The main remaining errors involve incorrect predicate-frame selection, argument-role confusion, hallucinated named entities, and failure on long sentences with multiple embedded events.

7 Discussion

We have introduced the first Abstract Meaning Representation dataset for the Konkani language, comprising 1,100 annotated sentences. We have established baselines for Konkani AMR parsing through zero-shot transfer ($F1 = 0.206$), supervised fine-tuning ($F1 = 0.331$), and language-adaptive pre-training followed by fine-tuning ($F1 = 0.437$). Each successive strategy yields a meaningful improvement, demonstrating that modest annotation investment combined with in-domain pre-training can produce increasingly capable low-resource AMR parsers. We release the dataset and all code to support continued research into semantic parsing for under-resourced South Asian languages.

Silver annotations are a viable bootstrapping strategy. The 88.4% human acceptance rate of Gemini 2.5 Pro annotations demonstrates that a capable LLM can produce high-quality AMR annotations for a low-resource language when given well-structured prompts, with good quality few shot examples. The fine-tuned model trained on these silver annotations achieves a Smatch $F1$ of 0.331, showing that silver-quality data is sufficient to bootstrap a working parser (after human verification and fixing of the data).

Language-adaptive pre-training provides consistent further gains. Exposing the model to a corpus of Konkani sentences before AMR fine-tuning increases Smatch $F1$ from 0.331 to 0.437. The gain in precision ($0.329 \rightarrow 0.500$) suggests that the pre-training stage improves the model’s ability to select correct concept frames and argument structures,

rather than just increasing output coverage. This result confirms that language-adaptive pre-training is a productive strategy for adapting multilingual seq2seq models to low-resource languages not covered by the original mBART model pre-training vocabulary.

Limitations. There are several limitations of this work. First is the dataset size, with 1,100 annotated sentences, our dataset is small compared to other standard AMR datasets. The limited training set means the fine-tuned parser has not seen many predicate types and is unlikely to generalise well to rare or semantically complex constructions. Second, there is no tokenisation support for Konkani in mBART, we use Hindi for tokenisation as a substitute, even though both have the same script they differ in semantic meaning. A tokeniser trained specifically on Konkani text would improve sub-word segmentation quality and likely boost downstream performance.

Future Directions Several directions emerge naturally from this work. Scaling the dataset to 5,000–10,000 sentences, potentially using other Konkani text sources (Wikipedia, court documents, literature) and the same LLM-annotation followed by human-verification pipeline would enable training significantly stronger parsers. Cross-lingual transfer from Indic languages might also help. Hindi and Marathi share substantial lexical overlap with Konkani. Pre-training or co-training on AMR data from these languages before fine-tuning on Konkani could provide additional signal. Adding Konkani specific vocabulary to the mBART tokeniser by training a sentencepiece model on our Konkani corpus and merging it could reduce tokenisation noise.

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A Appendix

A.1 Gemini Prompt

Below is the gemini prompt used for AMR annotation.

You are an expert linguist specializing in Abstract Meaning Representation (AMR).

AMR RULES (PENMAN notation):

STRUCTURE:

- Rooted, directed acyclic graph: (variable / concept :role value ...)
- Each node gets a unique short variable (b, b2, g, etc.)
- Co-reference: define a node once as (x / concept), then reuse x bare

CONCEPTS:

- Use PropBank verb frames with sense numbers: go-02, want-01, believe-01
- Nouns, adjectives, and verbs all become concepts (no POS distinction)
- Named entities: (p / person :name (n / name :op1 "John")) with :wiki if notable

CORE ROLES (ARGx are predicate-specific per PropBank):

- :ARG0 = agent/doer, :ARG1 = patient/theme, :ARG2 = beneficiary/instrument/etc
- Check PropBank frames for exact ARG meanings per predicate

NON-CORE ROLES:

- :location, :time, :manner, :purpose, :cause, :condition, :concession,
- :beneficiary, :instrument, :duration, :frequency, :degree,
- :source, :destination, :path, :topic, :medium

MODIFIERS & SPECIAL:

- Negation: :polarity -
- Modality: (o / obligate-01 :ARG2 ...) or (p / permit -01 ...)
- Questions: :polarity ? or (a / amr-unknown) for wh-questions
- Imperatives: :mode imperative
- Possession: :poss
- Part-whole: :part-of
- Degree: :degree (m / more), :degree (s / superlative)

CONJUNCTIONS & LISTS:

- (a / and :op1 ... :op2 ...) also or, contrast-01, etc

QUANTITIES & ENTITIES:

- Numbers: (x / concept :quant 3)
- Dates: (d / date-entity :year 2024 :month 3 :day 5)
- Percentages: (p / percentage -entity :value 25)

KEY ABSTRACTIONS (ignore English surface form):

- Drop articles, tense, aspect, passive voice
- "The girl adjusted the machine" = "The girl made adjustments to the machine" same AMR
- Inverse roles: use :ARG0-of, :ARG1-of for relative clauses and nominalizations

e.g. (d / doctor :ARG0-of (h
/ heal-01)) = "the
healing doctor"

- **Training time:** approximately 1.5 hours

Task: Given a Konkani sentence
, do the following:

1. Provide an accurate English translation.
2. Produce the AMR in valid PENMAN notation.

Output format (STRICT JSON
ONLY no markdown, no
explanation):
{{"english_translation":
"...","amr_penman":
"..."}}

Rules:

- Output ONLY valid JSON, no
markdown fences (Do NOT
use ```json)
- AMR must be valid PENMAN
using the rules above

Sentence: "{sentence}"

A.2 Model Hypterparamters

A.2.1 Supervised Fine-Tuning

- **Epochs:** 20
- **Batch size:** 4 per device, gradient accumula-
tion steps 4 (effective batch size 16)
- **Optimiser:** AdamW, learning rate 5×10^{-5} ,
weight decay 0.01
- **Warmup:** 100 steps
- **Max source length:** 128 tokens; **max target
length:** 512 tokens
- **Mixed precision:** fp16 on GPU

A.2.2 Pretraining + Fine-Tuning

- **Batch size:** 16 per device, gradient accumula-
tion steps 4 (effective batch size 64)
- **Optimiser:** AdamW, learning rate 3×10^{-5} ,
weight decay 0.01
- **Warmup:** 6% of total steps
- **Mixed precision:** fp16 on GPU (A100
80GB)